

Module 13 – Agile Methodology

Student Learning Notes

These notes are written based on my understanding after studying the Agile Methodology module. I have explained each topic in simple words as I understood it.

1. What I Studied in this Module

During this module I learned the basics of Agile software development, why companies prefer Agile over the traditional Waterfall model, how Scrum teams work, how projects are planned in sprints, how estimation is done using Story Points and Planning Poker, and how to write good User Stories with Acceptance Criteria.

2. Introduction to Agile

Agile is a software development methodology where work is completed in small iterations called sprints. Instead of building the complete software at once, the team develops small working features, collects feedback from customers, and improves the product continuously.

3. Agile Manifesto

Agile Manifesto contains 4 values and 12 principles. The main idea is to value people, collaboration, working software, customer feedback and responding to changes instead of following a rigid plan.

4. Agile vs Waterfall

Waterfall follows one phase after another and changes are difficult after development starts. Agile is flexible, welcomes changes, delivers software frequently, and involves customers throughout the project.

5. Scrum Framework

Scrum is the most popular Agile framework.

Roles:

- Product Owner – decides product requirements.
- Scrum Master – helps the team follow Scrum.
- Development Team – designs, develops and tests the product.

Ceremonies:

- Sprint Planning
- Daily Scrum
- Sprint Review
- Sprint Retrospective

Artifacts:

- Product Backlog
- Sprint Backlog
- Increment
- Definition of Done

6. Agile Estimation

Story Points are used to estimate effort instead of hours. Planning

Poker allows team members to estimate together using cards.

Velocity shows how much work a team completes in one sprint.

Burndown Chart tracks remaining work during the sprint.

7. User Stories

A User Story describes a feature from the user's perspective.

Format:

As a <user>, I want <goal>, so that <benefit>.

Example:

As a student, I want to view my attendance, so that I can monitor my academic progress.

8. INVEST Principle

Good user stories should be Independent, Negotiable, Valuable, Estimable, Small and Testable.

9. Acceptance Criteria

Acceptance Criteria clearly define when a feature is complete.

Example:

Given the user is on the login page,
When valid username and password are entered,
Then the dashboard should open successfully.

10. My Understanding

I understood that Agile focuses on teamwork, continuous improvement and customer satisfaction. Scrum provides a structured way to organize work through sprints. Estimation techniques help teams plan realistically. Writing clear User Stories and Acceptance Criteria reduces confusion during development.

11. Conclusion

This module helped me understand how modern software companies manage projects using Agile. I learned the importance of communication, iterative development and delivering value regularly.